

Short-Term Reversal Persists Globally—If Properly Measured

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Abstract

Short-term reversal appears to have weakened in international equity markets because standard measures mix firm-specific and industry-wide returns. When evaluated relative to industry peers, the reversal persists. Using data from 64 countries, the conventional strategy delivers negligible returns, whereas an industry-adjusted strategy earns 0.53% per month with a Sharpe ratio of 0.74. The effect reflects the reversal of firm-specific shocks masked by persistent industry components. A regret-based signal captures the same industry-relative variation and adds no independent return. Reversal did not disappear; it was mismeasured.

Keywords: short-term reversal, industry-adjusted reversal, intra-industry reversal, regret anomaly, international equity markets, return predictability, cross-section of stock returns

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1. Introduction

Short-term reversal once ranked among the most reliable cross-sectional patterns in equity returns (Jegadeesh, 1990; Lehmann, 1990). Recent evidence paints a different picture. In many markets—especially large and liquid ones—the effect appears to have weakened or even disappeared (McLean & Pontiff, 2016; Chordia et al., 2014; Chen & Velikov, 2023; Blitz et al., 2024). This shift raises a central question: has short-term reversal vanished, or do standard measures fail to detect it?

We show that the apparent decline reflects measurement, not economic decay. When recent returns are evaluated relative to industry peers, short-term reversal remains economically large and statistically significant across global equity markets. In contrast, the conventional signal based on raw returns delivers negligible returns. Raw returns combine firm-specific movements and industry-wide shocks. Firm-specific shocks tend to reverse, while industry components often persist (Da et al., 2014; Hameed & Mian, 2015; Blitz et al., 2024). When these forces offset each other, the standard signal attenuates predictability. Removing the industry component restores the effect.

We test this idea using monthly stock-level data from 64 countries over 1990–2023. We compare three signals that differ only in how they benchmark recent performance: (i) standard reversal based on raw returns, (ii) industry-adjusted reversal based on deviations from industry means, and (iii) a maximum-based signal motivated by the regret anomaly of Arisoy et al. (2024), defined relative to the best-performing peer within the same industry. The last two signals, therefore, represent two ways to improve the standard reversal measure using industry-relative performance. We form long–short portfolios based on each signal and evaluate their performance using standard return and alpha measures. We then examine whether the regret-based signal captures distinct information or a narrower behavioral version of the broader industry-adjusted effect.

The evidence is clear. In the pooled global sample, the standard reversal strategy earns 0.05% per month with an insignificant alpha, while the industry-adjusted strategy earns 0.53% per month, with a Sharpe ratio of 0.74 and a six-factor alpha of 0.60% ($t = 4.14$). The contrast is strongest in large markets. In the United Kingdom and China, standard reversal yields negligible returns, whereas the industry-adjusted strategy earns 0.87% and 0.99% per month, respectively. Figure 1 illustrates the difference.

[Insert Figure 1 here]

This framework also clarifies the relationship between industry-adjusted reversal and the regret-based signal of Arisoy et al. (2024). Both signals rank stocks by relative underperformance within industries, but they use different benchmarks: the industry mean in one case and the industry winner in the other. In the global sample, the regret

strategy earns 0.40% per month, indicating that a behavioral, peer-based refinement also improves upon the standard reversal signal. Yet once we control for industry-adjusted reversal, the regret alpha falls to 0.25% ($t = 1.34$). The evidence, therefore, points to a common underlying effect: the regret signal does not add independent information; it reflects the same cross-sectional variation captured by industry-adjusted reversal.

Related literature. Our paper relates to the short-term reversal literature initiated by Jegadeesh (1990) and Lehmann (1990). Reversal has been studied extensively in U.S. markets (Nagel, 2012; Hameed & Mian, 2015; Blitz et al., 2024), internationally (Jegadeesh et al., 2025), and in specific settings such as China (Chen et al., 2022; Neszveda et al., 2022) and international markets (Butt et al., 2021, 2024; Butt & Sadaqat, 2024; Kang et al., 2025). Prior work links reversal to liquidity provision (Campbell et al., 1993; Avramov et al., 2006; Cheng et al., 2017; Da et al., 2014) and behavioral mechanisms, including overreaction (De Bondt & Thaler, 1985; Heyman et al., 2019), sentiment (Daniel et al., 1998; Dyl et al., 2019), and under-inference (Farkas & Neszveda, 2025). We build most directly on Da et al. (2014) and Blitz et al. (2024), extending their industry-based approach to 64 international markets.

Furthermore, our evidence speaks to anomaly attenuation (McLean & Pontiff, 2016; Chen & Zimmermann, 2022; Chordia et al., 2014; Chen & Velikov, 2023; Gordon et al., 2025; Brogaard & Zarei, 2023). We relate most closely to international studies (Jacobs & Müller, 2020; Hollstein, 2022) and are the first to focus on short-term reversal in this context. Our results align with Jacobs and Müller (2020), showing that anomalies need not disappear when carefully measured.

Finally, we contribute to the literature on the regret anomaly (Arisoy et al., 2024), studied in the U.S. and Japan (Kim & Ohk, 2025), China (Wang et al., 2026), and globally (Atilgan et al., 2025). Our evidence places that anomaly in a broader framework: regret-based reversal also improves on the raw reversal signal because it evaluates stocks relative to industry peers, but across 64 markets, its return largely reflects the same industry-relative variation captured by the more general industry-adjusted measure.

The remainder of the paper proceeds as follows. Section 2 describes the data and methods. Section 3 presents the findings. Finally, Section 4 concludes.

2. Data and Methods

We use monthly stock-level data for common equities across 64 developed and emerging markets from January 1990 to December 2023. For each stock-month, the dataset includes excess returns, market capitalization, industry identifiers used to construct relative variables, firm characteristics for factor construction, lagged returns, and industry-adjusted performance measures. U.S. data are from CRSP, while international

return and accounting data come from Compustat. We follow Jensen et al. (2023) for data construction and express all variables in U.S. dollars using Compustat exchange rates, consistent with standard practice in international asset pricing (e.g., Fama and French, 2012, 2017).

We restrict the sample to common stocks listed on the main exchanges. At the stock level, we exclude observations with missing returns, prices ≤ 1 , missing industry codes, and nano-cap firms. At the country-month level, we require at least 10 stocks per market. Table A1 in the Online Appendix reports the cross-country composition. The sample includes between 7,206 and 21,698 stocks, with an average of 14,193, and a total of 5,790,595 monthly observations. The number and aggregate market capitalization of firms increase over time, as shown in Figure 2.

The first signal is the standard short-term reversal measure used in the literature (e.g., Jegadeesh, 1990; Lehmann, 1990). The second is the industry-adjusted reversal signal, defined as the deviation of a stock's return from the mean return of its industry peers over the previous month,

$$REV_{i,t}^{IN} = R_{i,t-1} - R_{j,t-1}, \quad (1)$$

where $R_{j,t-1}$ denotes the average return in industry j . The third signal follows the regret-based approach of Arisoy et al. (2024) and measures the deviation from the best-performing stock in the industry:

$$REG_{i,t} = R_{i,t-1} - \max_{k \in j}(R_{k,t-1}), \quad (2)$$

where $\max_{k \in j}(R_{k,t-1})$ is the return of the top-performing stock in industry j .

Within each country and month, stocks are sorted into quintile portfolios based on each signal. We compute equal- and value-weighted returns. In all cases, the strategy takes long positions in stocks with the strongest relative underperformance and short positions with the strongest relative outperformance.

3. Results

This section presents the main results. We first compare standard and industry-adjusted reversal, then examine robustness and scrutinize the link between intra-industry reversal and the regret-anomaly-based refinement.

3.1. Baseline Findings

Table 1 reports the performance of standard and industry-adjusted short-term reversal across 64 equity markets and in the pooled global sample. The contrast is remarkable. The standard reversal strategy yields weak, often insignificant results, whereas the industry-adjusted alternative remains economically large and statistically significant, consistent with U.S. evidence (Blitz et al., 2024) and extending it internationally.

[Insert Table 1 here]

In the pooled global sample, the standard strategy generates an average monthly return of 0.05% with a six-factor alpha of 0.18%, neither statistically significant. At the country level, significant returns appear in only 13 of 64 markets, mostly smaller or less liquid economies such as Chile, Peru, and the Philippines. In large markets, performance remains weak: 0.01% per month in the United States, 0.15% in the United Kingdom, and 0.16% in Germany, none statistically significant.

By contrast, the industry-adjusted strategy earns 0.53% per month, with a Sharpe ratio of 0.74 and a six-factor alpha of 0.60% ($t = 5.02$ and 4.14). This represents a substantial improvement over the standard signal. The strategy produces statistically significant returns and alphas in 22 of 64 markets, including large and liquid economies such as France (0.68%), Japan (0.82%), the United Kingdom (0.87%), and the United States (0.34%).

3.2. Robustness checks

Table 2 examines robustness across time and firm-size groups. The industry-adjusted strategy delivers positive and statistically significant returns across subperiods, micro-, small-, and large-cap stocks, and in the global equal-weighted sample, while the standard strategy remains weak.

[Insert Table 2 here]

Performance is stronger in the first half of the sample but remains economically and statistically significant in the second, indicating attenuation rather than disappearance. Returns are higher among micro and small caps, yet remain significant for large firms, suggesting the effect does not rely on a single segment. Equal-weighted results point in the same direction. Table 2 reports global equal-weighted portfolios, while detailed country-level results appear in Table A2 in the Online Appendix. Overall, the advantage of the industry-adjusted strategy holds across periods, firm sizes, and weighting schemes, indicating that short-term reversal persists once returns are measured relative to industry peers.

3.3. Regret-Based Reversal as an Industry-Relative Refinement

Arisoy et al. (2024) construct a regret-based signal as the deviation between a stock’s return and the maximum return within its industry. That signal fits naturally into our framework. Both regret-based reversal and industry-adjusted reversal improve on the standard specification by evaluating recent performance relative to industry peers rather than in isolation. They differ only in the benchmark: the industry-adjusted signal uses the peer average, whereas the regret signal uses the industry winner. Implicitly, regret may capture a sharper, behaviorally motivated version of the same underlying industry-relative reversal.

Table 3 supports this interpretation. Panel A shows that the regret strategy is economically meaningful on its own. In the pooled global sample, it earns 0.40% per month with an alpha of 0.39%. Several markets exhibit significant returns, although the effect is less widespread than for the industry-adjusted strategy. Benchmarking stocks against industry peers, whether using the mean or the best performer, improves the raw reversal signal.

[Insert Table 3 here]

Panel B sharpens the distinction between the two signals. If regret captured an independent return premium, the intercept would remain large after controlling for the industry-adjusted strategy. It does not. In the pooled sample, the spanning alpha falls to 0.25% ($t = 1.34$), while the loading on industry-adjusted reversal remains positive and significant, and the regression delivers an R^2 of 0.05, with higher explanatory power in several individual markets. The regret signal does not introduce a separate source of predictability beyond the industry-adjusted measure. Instead, regret returns co-move with industry-adjusted reversal, consistent with both signals ranking stocks on the same industry-relative dimension, with the regret measure placing more weight on extreme peer comparisons.

The cross-country evidence corroborates these conclusions. Only 5 markets exhibit a significant spanning alpha, whereas most markets show insignificant intercepts once industry-adjusted reversal is included. By contrast, exposure to the industry-adjusted strategy remains strong across a broad set of countries. Additional tests in the Online Appendix confirm the same pattern. Equal-weighted results are similar (Table A3), bidirectional spanning tests show that industry-adjusted reversal subsumes the regret strategy whereas the reverse does not (Table A4), and the relation persists across subperiods and firm-size groups (Table A5).

Our results place the regret premium inside the broader industry-adjusted reversal effect. Regret-based reversal improves on the raw signal by evaluating stocks against industry

peers, but it does not stand apart from the more general industry-adjusted measure. The industry-adjusted specification remains our primary measure, while the regret signal offers a behavioral interpretation built around extreme peer comparisons.

4. Concluding Remarks

Industry-adjusted short-term reversal remains strong across international markets, whereas the standard measure delivers weak results. The effect survives across subperiods and specifications. A regret-based signal also improves on the standard reversal measure by evaluating stocks relative to industry peers, but its return largely reflects the broader industry-adjusted effect. The apparent decline of short-term reversal, therefore, reflects measurement, not economic decay.

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Figure 1. Cumulative Returns of Short-Term Reversal and Related Strategies.

The figure plots cumulative returns of three strategies using monthly data from 64 countries over 1990–2023. The signals differ only in how they measure recent performance: (i) standard reversal based on raw returns, (ii) industry-adjusted reversal based on deviations from industry means, and (iii) a maximum-based signal defined relative to the best-performing peer. Each strategy forms value-weighted long–short quintile portfolios, buying (selling) the worst (best) past performers.

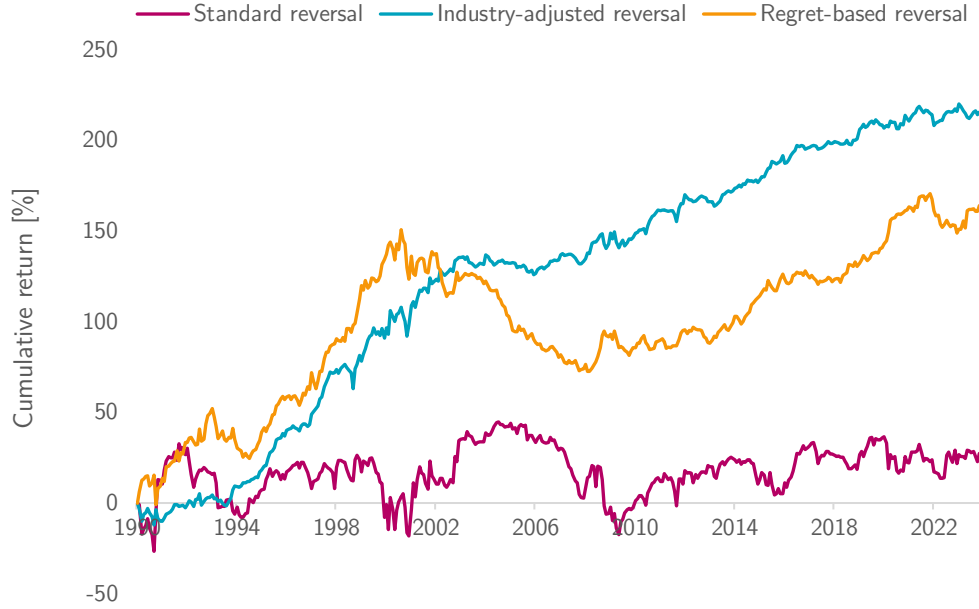
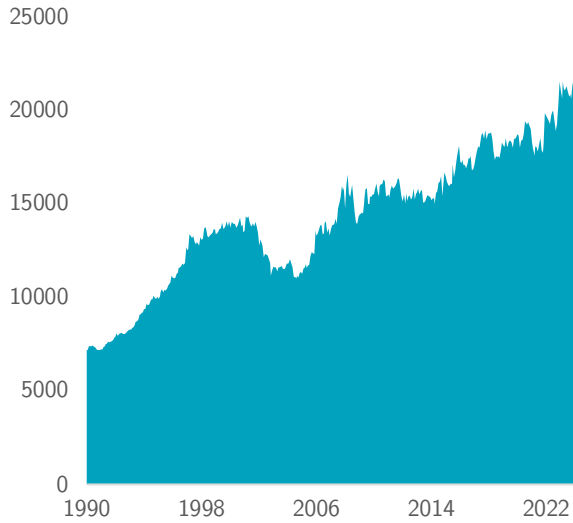


Figure 2. Sample Size over Time

The figure presents the size of the research sample over time in terms of the number of stocks (Panel A) and their total market capitalization in USD billion (Panel B). The sample consists of 64 countries, and the whole study period covers January 1990 to December 2023.

Panel A: Number of stocks



Panel B: Total market capitalization (USD billion)

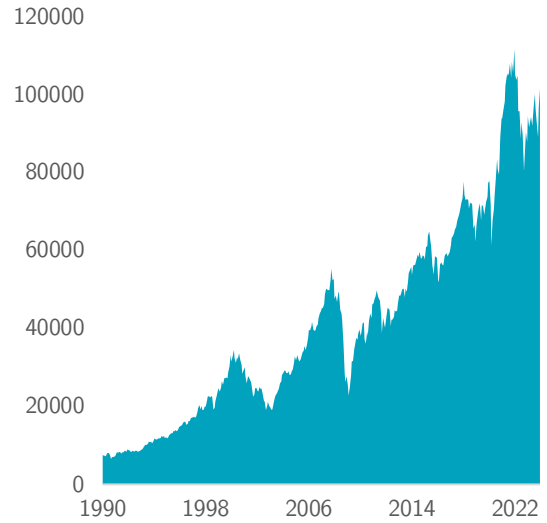


Table 1. Short-Term Reversal Strategies Across International Equity Markets

The table reports the performance of standard and industry-adjusted short-term reversal strategies across international equity markets. The signals differ in how they measure recent performance: the standard strategy uses raw past returns, while the industry-adjusted strategy removes industry effects. In both cases, long–short portfolios buy (sell) value-weighted quintiles of stocks with the strongest relative underperformance (outperformance). R denotes the mean monthly return, SR the annualized Sharpe ratio, and α the monthly alpha relative to the Fama–French six-factor model. Newey–West (1987) t-statistics are reported in parentheses. The last row (“# |t-stat| > 1.96”) reports the number of markets with statistically significant estimates. Returns and alphas are in percent. The sample period is January 1990 to December 2023.

	Industry-adjusted short-term reversal			Standard short-term reversal		
	R	SR	α	R	SR	α
Argentina	1.19 (2.52)	0.50	1.14 (1.90)	0.54 (0.79)	0.15	1.00 (1.22)
Australia	0.28 (1.81)	0.29	0.23 (1.55)	0.04 (0.18)	0.03	0.10 (0.41)
Austria	-0.09 (-0.37)	-0.06	-0.07 (-0.27)	0.28 (0.88)	0.15	0.39 (1.21)
Bangladesh	-0.56 (-1.35)	-0.28	-0.08 (-0.28)	-0.99 (-1.51)	-0.37	-0.26 (-0.46)
Belgium	0.05 (0.23)	0.03	-0.11 (-0.45)	-0.27 (-0.84)	-0.15	-0.46 (-1.25)
Brazil	0.19 (0.59)	0.13	0.23 (0.64)	-0.91 (-2.07)	-0.45	-1.07 (-2.25)
Bulgaria	-0.25 (-0.62)	-0.15	-0.02 (-0.06)	0.06 (0.09)	0.02	-0.14 (-0.15)
Canada	0.33 (1.41)	0.17	0.23 (0.80)	0.04 (0.14)	0.02	0.11 (0.33)
Chile	0.58 (2.33)	0.42	0.51 (2.23)	0.70 (2.38)	0.38	0.60 (2.21)
China	0.99 (3.73)	0.76	0.64 (2.92)	0.58 (1.52)	0.31	0.23 (0.80)
Colombia	0.63 (2.23)	0.42	0.36 (1.10)	-0.03 (-0.09)	-0.01	-0.14 (-0.26)
Côte d’Ivoire	0.58 (1.04)	0.30	-0.25 (-0.46)	1.24 (2.71)	0.46	0.94 (1.64)
Croatia	-0.01 (-0.02)	-0.01	-0.36 (-0.84)	0.46 (0.96)	0.24	0.07 (0.12)
Czech Republic	1.19 (2.59)	0.52	2.07 (2.36)	0.31 (0.63)	0.09	-0.10 (-0.10)
Denmark	-0.04 (-0.16)	-0.03	0.06 (0.21)	-0.37 (-1.36)	-0.20	-0.03 (-0.08)
Egypt	-0.11 (-0.25)	-0.05	0.45 (0.90)	-0.52 (-0.87)	-0.19	-0.06 (-0.09)
Finland	0.10 (0.25)	0.05	0.29 (0.68)	-0.52 (-1.44)	-0.22	-0.65 (-1.73)
France	0.68 (3.46)	0.59	0.62 (2.75)	0.56 (2.00)	0.37	0.41 (1.49)
Germany	0.36 (1.91)	0.27	0.49 (2.15)	0.16 (0.63)	0.09	0.46 (1.70)
Greece	0.53 (1.31)	0.26	0.92 (2.42)	0.24 (0.51)	0.09	0.47 (0.97)
Hong Kong	0.30 (1.22)	0.19	0.42 (1.70)	0.05 (0.14)	0.02	0.23 (0.62)
Hungary	0.09 (0.20)	0.04	0.72 (1.17)	-0.02 (-0.04)	-0.01	-0.41 (-0.67)
India	0.63 (2.93)	0.37	0.63 (2.15)	-0.30 (-0.67)	-0.12	0.25 (0.49)
Indonesia	-0.21 (-0.42)	-0.09	-0.34 (-0.47)	-0.41 (-0.61)	-0.14	-0.02 (-0.02)
Ireland	0.22 (0.55)	0.10	0.07 (0.17)	-0.27 (-0.48)	-0.09	-0.43 (-0.73)
Israel	0.55 (1.88)	0.33	0.32 (0.89)	-0.31 (-0.89)	-0.15	-0.65 (-1.59)
Italy	-0.08 (-0.35)	-0.06	-0.16 (-0.63)	-0.37 (-1.49)	-0.22	-0.17 (-0.66)
Japan	0.82 (4.63)	0.70	0.84 (4.43)	0.94 (4.03)	0.63	0.98 (3.74)
Jordan	0.02 (0.05)	0.01	-0.42 (-0.94)	-1.13 (-2.11)	-0.53	-0.93 (-1.88)
Kuwait	0.48 (1.82)	0.33	0.39 (1.34)	0.40 (1.05)	0.24	0.25 (0.64)
Luxembourg	0.23 (0.33)	0.11	3.08 (4.08)	1.45 (1.55)	0.50	2.71 (1.33)
Malaysia	0.57 (2.26)	0.42	0.49 (2.37)	-0.07 (-0.24)	-0.04	-0.01 (-0.02)
Mauritius	0.21 (0.87)	0.21	0.08 (0.27)	-0.28 (-0.85)	-0.17	0.17 (0.49)
Mexico	0.07 (0.26)	0.05	0.04 (0.12)	0.75 (2.30)	0.43	0.47 (1.46)
Morocco	0.51 (1.91)	0.43	0.60 (1.98)	0.70 (1.55)	0.40	0.86 (2.38)
Netherlands	0.20 (1.05)	0.14	0.04 (0.16)	0.04 (0.12)	0.02	-0.22 (-0.65)
New Zealand	0.27 (1.12)	0.20	0.25 (1.03)	-0.40 (-1.36)	-0.24	-0.22 (-0.70)
Norway	0.10 (0.29)	0.05	0.13 (0.43)	-0.16 (-0.46)	-0.08	-0.16 (-0.47)
Oman	-0.51 (-1.16)	-0.27	-0.34 (-0.86)	-1.20 (-2.71)	-0.74	-1.15 (-2.73)
Pakistan	0.98 (2.57)	0.53	1.24 (3.58)	0.99 (1.95)	0.41	1.05 (2.02)

Peru	1.49 (3.84)	0.81	1.40 (3.33)	1.19 (2.50)	0.54	1.55 (2.58)
Philippines	1.74 (3.47)	0.91	1.17 (4.55)	1.47 (2.44)	0.54	1.03 (2.27)
Poland	0.83 (3.11)	0.58	0.86 (2.59)	-0.60 (-1.36)	-0.28	-0.26 (-0.68)
Portugal	0.03 (0.09)	0.02	0.16 (0.50)	0.08 (0.17)	0.03	0.27 (0.65)
Qatar	0.81 (2.48)	0.50	0.97 (2.51)	0.92 (1.78)	0.44	0.97 (1.99)
Russia	0.61 (1.13)	0.26	0.59 (1.04)	0.95 (1.24)	0.34	0.47 (0.62)
Saudi Arabia	1.23 (4.84)	0.93	1.11 (3.62)	-0.38 (-0.86)	-0.17	-0.18 (-0.41)
Singapore	0.25 (1.36)	0.21	0.13 (0.68)	0.01 (0.04)	0.01	-0.27 (-0.87)
Slovenia	-0.40 (-1.10)	-0.26	0.33 (0.56)	0.40 (0.68)	0.19	-0.25 (-0.39)
South Africa	1.10 (5.38)	0.89	1.10 (5.03)	0.12 (0.35)	0.06	0.35 (0.97)
South Korea	0.44 (1.81)	0.25	0.40 (1.29)	0.39 (0.94)	0.16	0.41 (0.82)
Spain	0.42 (1.76)	0.30	0.25 (0.81)	0.20 (0.76)	0.12	0.20 (0.67)
Sri Lanka	0.47 (0.99)	0.28	0.55 (1.09)	-0.46 (-0.66)	-0.17	0.47 (0.58)
Sweden	0.55 (2.57)	0.39	0.43 (2.04)	0.52 (1.48)	0.26	0.51 (1.51)
Switzerland	0.51 (2.70)	0.36	0.53 (2.87)	0.23 (1.12)	0.14	0.45 (1.81)
Taiwan	0.25 (0.92)	0.16	0.25 (0.79)	-0.16 (-0.39)	-0.07	-0.18 (-0.44)
Thailand	0.42 (1.27)	0.22	0.29 (0.97)	-0.83 (-2.13)	-0.34	-0.89 (-2.54)
Trinidad and Tobago	-0.54 (-2.10)	-0.57	-0.37 (-1.66)	-1.37 (-3.48)	-1.02	-1.13 (-2.82)
Tunisia	0.21 (0.65)	0.16	0.37 (1.19)	-0.38 (-1.01)	-0.26	0.02 (0.06)
Turkey	-0.05 (-0.06)	-0.02	-0.57 (-0.67)	0.24 (0.34)	0.07	-0.51 (-0.57)
UAE	1.73 (3.51)	0.82	1.40 (2.79)	0.16 (0.23)	0.06	0.21 (0.28)
United Kingdom	0.87 (5.37)	0.82	0.86 (4.83)	0.15 (0.57)	0.10	-0.06 (-0.19)
United States	0.34 (2.12)	0.32	0.30 (1.87)	-0.01 (-0.05)	-0.01	0.06 (0.29)
Vietnam	-0.11 (-0.22)	-0.05	-0.37 (-0.70)	-1.41 (-3.58)	-0.61	-1.81 (-3.05)
Global (pooled)	0.53 (5.02)	0.74	0.60 (4.14)	0.05 (0.24)	0.04	0.18 (0.66)
# t-stat > 1.96	23		23	13		12

Table 2. Portfolio Performance Across Time and Firm Size

The table reports the mean monthly returns, Sharpe ratios, and alphas relative to the Fama-French six-factor model on global portfolios tested on different samples, as indicated in the first row. These include time-and size-specific samples of value-weighted portfolios, as well as the full, equal-weighted sample. An industry-adjusted short-term reversal strategy follows the industry-relative formulation, in which stock returns are adjusted for industry effects before constructing the reversal signal. The long-short portfolios buy (sell) a value-weighted (equal-weighted) quintile of stocks with the highest (lowest) relative underperformance (overperformance). The standard short-term reversal strategy is based on raw past returns, where the long-short portfolios buy (sell), a value-weighted (equal-weighted) quintile of stocks with the lowest (highest) recent returns. R is the mean monthly return, SR is the annualized Sharpe ratio, and α represents the monthly alpha relative to the Fama-French six-factor model. t-statistics are reported in parentheses, while the last row reports the number of markets with $|t\text{-statistic}| > 1.96$. R and α are expressed in percent. The full study period is January 1990 – December 2023.

	Industry-adjusted short-term reversal			Standard short-term reversal		
	R	SR	α	R	SR	α
First subperiod 1990-2006	0.67 (3.92)	0.79	0.87 (3.62)	0.15 (0.46)	0.09	0.62 (1.43)
Second subperiod 2007-2023	0.40 (3.15)	0.69	0.38 (2.59)	-0.05 (-0.18)	-0.04	-0.25 (-0.83)
Micro caps	1.03 (7.55)	1.35	1.40 (6.54)	-0.39 (-1.89)	-0.30	-0.06 (-0.20)
Small caps	0.78 (8.28)	1.28	0.76 (6.89)	-0.27 (-1.23)	-0.20	-0.40 (-1.54)
Large caps	0.43 (4.38)	0.61	0.41 (3.57)	0.15 (0.72)	0.11	0.19 (0.73)
Global (equal-weighted)	0.91 (8.46)	1.43	0.93 (6.32)	-0.24 (-1.24)	-0.19	-0.21 (-0.86)

Table 3. Regret Anomaly and Spanning Against Industry-Adjusted Short-Term Reversal

The table reports the performance of the regret strategy and the results of time-series spanning regressions estimated separately for each market over the full sample period. The regret strategy follows Arisoy et al. (2024): long–short portfolios buy (sell) value-weighted quintiles of stocks with the strongest underperformance (outperformance) relative to the best-performing stock within the same industry. Panel B reports spanning regressions of regret returns on industry-adjusted short-term reversal:

$$R_{i,t}^{REG} = \alpha_{REG|REV^{IN},i} + \beta_{REV^{IN},i} R_{i,t}^{REV^{IN}} + \varepsilon_{i,t},$$

where $R_{i,t}^{REV^{IN}}$ denotes the return of the industry-adjusted reversal portfolio. The intercept $\alpha_{REG|REV^{IN},i}$ captures the component of regret returns not explained by industry-adjusted reversal, and $\beta_{REV^{IN},i}$ measures exposure to the industry-adjusted strategy. The last column reports adjusted R^2 . R denotes the mean monthly return, SR the annualized Sharpe ratio, and α the monthly alpha relative to the Fama–French six-factor model. Newey–West (1987) t-statistics are reported in parentheses. The last row (“# |t-stat| > 1.96”) reports the number of markets with statistically significant estimates. Returns and alphas are in percent. The sample period is January 1990 to December 2023.

	Panel A: Regret anomaly			Panel B: Spanning vs industry-adjusted short-term reversal		
	R	SR	α	$\alpha_{REG REV^{IN}}$	$\beta_{REV^{IN}}$	R^2
Argentina	1.85 (3.47)	0.64	1.70 (2.29)	1.17 (2.38)	0.30 (3.12)	0.07
Australia	0.15 (0.71)	0.13	-0.08 (-0.38)	-0.05 (-0.25)	0.48 (8.66)	0.17
Austria	0.04 (0.18)	0.02	0.00 (-0.02)	-0.09 (-0.32)	0.56 (7.46)	0.27
Bangladesh	0.16 (0.41)	0.09	0.33 (1.02)	0.03 (0.10)	0.37 (6.90)	0.16
Belgium	0.51 (2.84)	0.35	0.38 (1.82)	0.60 (2.26)	0.48 (9.70)	0.22
Brazil	0.13 (0.39)	0.09	0.31 (0.93)	0.00 (0.01)	0.44 (4.41)	0.17
Bulgaria	-0.27 (-0.41)	-0.11	-0.36 (-0.54)	-0.13 (-0.24)	0.52 (2.82)	0.16
Canada	-0.21 (-0.57)	-0.09	-0.15 (-0.41)	-0.17 (-0.56)	0.35 (2.33)	0.12
Chile	0.55 (2.13)	0.35	0.21 (0.71)	0.32 (1.02)	0.49 (8.98)	0.22
China	0.38 (1.64)	0.29	0.37 (1.68)	-0.01 (-0.05)	0.38 (5.10)	0.16
Colombia	0.36 (0.96)	0.18	-0.17 (-0.45)	-0.25 (-0.77)	0.59 (6.35)	0.20
Côte d'Ivoire	0.67 (1.18)	0.32	-0.02 (-0.03)	0.58 (1.31)	0.32 (3.29)	0.10
Croatia	-0.26 (-0.70)	-0.16	-0.41 (-1.43)	0.04 (0.15)	0.54 (6.00)	0.35
Czech Republic	1.31 (2.12)	0.54	1.49 (1.60)	0.67 (1.27)	0.32 (1.01)	0.04
Denmark	0.15 (0.52)	0.09	0.23 (0.85)	-0.01 (-0.05)	0.40 (5.52)	0.15
Egypt	0.60 (1.24)	0.25	1.31 (2.90)	0.42 (1.06)	0.42 (4.40)	0.18
Finland	0.02 (0.07)	0.01	0.12 (0.41)	-0.38 (-1.14)	0.49 (7.50)	0.26
France	0.30 (1.66)	0.27	0.47 (2.34)	0.03 (0.15)	0.40 (7.19)	0.18
Germany	0.36 (1.71)	0.28	0.81 (3.32)	0.05 (0.28)	0.50 (7.18)	0.24
Greece	0.02 (0.05)	0.01	0.72 (1.85)	-0.53 (-1.15)	0.67 (7.33)	0.28
Hong Kong	-0.23 (-0.86)	-0.15	-0.22 (-0.83)	-0.18 (-0.85)	0.49 (6.80)	0.26
Hungary	-0.08 (-0.17)	-0.03	-0.10 (-0.14)	-0.34 (-0.67)	0.09 (1.14)	0.01
India	0.40 (1.44)	0.20	0.28 (1.12)	0.44 (2.18)	0.50 (8.22)	0.23
Indonesia	-0.83 (-1.53)	-0.32	-0.28 (-0.39)	0.70 (1.54)	0.58 (5.73)	0.30
Ireland	0.28 (0.65)	0.12	0.56 (1.36)	-0.15 (-0.38)	0.48 (5.29)	0.20
Israel	0.00 (0.00)	0.00	0.01 (0.02)	-0.31 (-1.23)	0.45 (8.80)	0.22
Italy	-0.36 (-1.33)	-0.27	-0.26 (-1.10)	-0.17 (-0.70)	0.52 (9.24)	0.19
Japan	0.43 (2.97)	0.48	0.63 (4.23)	0.09 (0.62)	0.41 (7.13)	0.30
Jordan	-0.60 (-1.11)	-0.29	-0.69 (-1.26)	-0.35 (-0.69)	0.59 (7.12)	0.26
Kuwait	0.22 (0.78)	0.14	0.30 (0.91)	0.31 (1.06)	0.62 (10.12)	0.33
Luxembourg	0.72 (1.32)	0.33	1.33 (1.00)	0.30 (0.64)	0.59 (7.54)	0.36
Malaysia	0.18 (0.82)	0.14	0.12 (0.56)	0.18 (0.74)	0.36 (3.78)	0.14

Mauritius	-0.67 (-2.04)	-0.56	-0.69 (-1.97)	-0.64 (-2.11)	0.24 (2.17)	0.04
Mexico	-0.09 (-0.35)	-0.06	-0.14 (-0.57)	0.01 (0.05)	0.49 (8.59)	0.23
Morocco	0.36 (1.42)	0.31	0.29 (1.12)	-0.03 (-0.09)	0.37 (5.73)	0.13
Netherlands	0.80 (3.31)	0.48	0.81 (2.67)	0.67 (2.91)	0.58 (6.51)	0.22
New Zealand	0.19 (0.81)	0.15	0.17 (0.69)	0.04 (0.18)	0.28 (4.23)	0.09
Norway	-0.37 (-1.07)	-0.20	-0.39 (-1.27)	-0.34 (-1.09)	0.46 (7.02)	0.20
Oman	-0.68 (-1.48)	-0.37	-0.47 (-1.19)	-0.39 (-1.09)	0.51 (3.31)	0.27
Pakistan	0.37 (0.70)	0.17	1.17 (3.10)	0.16 (0.32)	0.51 (5.13)	0.18
Peru	0.83 (1.77)	0.38	0.46 (0.80)	-0.03 (-0.06)	0.62 (9.97)	0.29
Philippines	1.19 (2.43)	0.49	0.69 (2.02)	-0.01 (-0.04)	0.67 (4.18)	0.28
Poland	0.55 (1.78)	0.32	0.46 (1.58)	0.01 (0.04)	0.48 (6.71)	0.16
Portugal	-0.06 (-0.19)	-0.03	-0.08 (-0.21)	0.54 (1.78)	0.38 (5.39)	0.13
Qatar	0.49 (1.14)	0.24	0.93 (2.31)	0.09 (0.27)	0.48 (6.18)	0.17
Russia	0.88 (1.81)	0.40	0.87 (1.66)	0.71 (1.93)	0.54 (8.16)	0.34
Saudi Arabia	0.62 (2.20)	0.35	0.82 (2.08)	-0.07 (-0.17)	0.62 (8.16)	0.21
Singapore	-0.11 (-0.39)	-0.08	-0.33 (-1.39)	-0.22 (-0.88)	0.35 (2.97)	0.09
Slovenia	0.14 (0.30)	0.07	0.69 (1.01)	-0.54 (-1.54)	0.09 (1.02)	0.01
South Africa	0.67 (2.31)	0.42	0.69 (2.34)	0.23 (0.85)	0.57 (8.26)	0.19
South Korea	0.68 (1.62)	0.32	0.95 (3.06)	0.28 (0.77)	0.44 (3.76)	0.12
Spain	0.07 (0.31)	0.04	0.19 (0.73)	-0.24 (-1.18)	0.57 (9.47)	0.31
Sri Lanka	0.75 (1.49)	0.40	0.92 (1.98)	-0.26 (-0.49)	0.38 (3.07)	0.11
Sweden	0.40 (1.40)	0.25	0.60 (2.04)	-0.04 (-0.12)	0.43 (6.58)	0.14
Switzerland	0.02 (0.08)	0.01	0.47 (2.19)	-0.17 (-0.87)	0.50 (8.24)	0.26
Taiwan	0.78 (2.08)	0.38	-0.07 (-0.20)	0.80 (1.93)	0.35 (4.68)	0.07
Thailand	0.66 (2.03)	0.32	0.61 (2.00)	0.16 (0.54)	0.64 (6.18)	0.34
Trinidad and Tobago	-0.27 (-1.14)	-0.28	-0.48 (-1.79)	0.04 (0.17)	0.52 (4.63)	0.24
Tunisia	0.20 (0.59)	0.15	0.37 (1.38)	0.14 (0.41)	0.40 (3.42)	0.16
Turkey	0.07 (0.14)	0.03	-0.41 (-0.73)	0.50 (1.39)	0.27 (2.92)	0.17
UAE	1.54 (2.58)	0.70	1.35 (2.08)	0.06 (0.10)	0.54 (13.28)	0.26
United Kingdom	0.49 (2.37)	0.44	0.62 (3.55)	-0.04 (-0.20)	0.50 (7.96)	0.24
United States	0.27 (1.38)	0.24	0.65 (3.19)	0.15 (0.63)	0.34 (4.56)	0.10
Vietnam	0.28 (0.53)	0.14	0.31 (0.60)	-0.48 (-1.18)	0.60 (10.22)	0.34
Global (pooled)	0.40 (2.37)	0.42	0.39 (2.03)	0.25 (1.34)	0.30 (2.36)	0.05
# t-stat > 1.96	15		21	5		